
Viscoelastic Fluids in Large Amplitude Oscillatory Shear (LAOS)

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Chapter 1

Background Theory

1.1 Introduction to Rheology and Non-Newtonian Fluids

The relationship between the stress applied to a material and its kinematic response is known as its rheology. Newtonian fluids are characterised by a Newtonian rheology, where there is a linear relation between stress and strain-rate. However, apart from extremely common fluids such as air and water, virtually no fluid is perfectly Newtonian. The classical Newtonian assumption breaks down for flows that alter the local microstructure of the fluid. Non-Newtonian fluids are an active field of research and have vast biological, industrial and geophysical applications. In the biological realm, blood is known to exhibit non-Newtonian behavior such as thixotropy and viscoelasticity (Beris et al., 2021). On the other hand, in the geophysical realm, mud and silicate lava are known to exhibit a yield stress (Balmforth and Craster, 2001).

1.2 Governing Equations

We first briefly describe the continuum approximation that underpins the derivation of the governing equations. Let L be a representative physical length scale of the system and let l be a representative length scale of the microstructure of the fluid (such as the mean free path length). Then we require that $L \gg l$. The length scale L will determine the size of a fluid element and we require that the volume ($\sim L^3$) is not too large, so that variations in material quantities are negligible within a fluid parcel and the variations between neighboring parcels are approximately continuous.

If the fluid is assumed to be incompressible, then mass conservation gives:

$$\nabla \cdot \mathbf{u} = 0, \quad (1.1)$$

where $\mathbf{u}$ denotes the Eulerian velocity field.

We denote σ_{ij} as the i,j -th component of the stress tensor of the fluid. A physical interpretation of σ_{ij} is that it represents the i -th component of stress on a surface element δS , which has normal $\mathbf{n}$ pointing in the j -direction. If we denote local stress $\mathbf{t}$ on a surface area element δS with outer unit normal $\mathbf{n}$ as $\mathbf{t}$, then the stress is given by the Cauchy Stress Theorem as:

$$t_i = \sigma_{ij} n_j. \quad (1.2)$$

We also note that as a consequence of the conservation of angular momentum, the stress tensor must be symmetric.

The conservation of momentum gives:

$$\rho \frac{D\mathbf{u}}{Dt} = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{F}, \quad (1.3)$$

where ρ is the fluid density, $D/Dt = \partial/\partial t + \mathbf{u} \cdot \nabla$ is the material derivative and $\mathbf{F}$ is the force per unit mass.

The stress tensor is decomposed into an isotropic part ($-p\mathbf{I}$) and a deviatoric part denoted $\boldsymbol{\tau}$. We may occasionally use $\boldsymbol{\sigma}$ to refer to the deviatoric stress and this will be made clear, based on the context.

This gives the following equation of motion:

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \rho \mathbf{F}. \quad (1.4)$$

To close the set of equations, we need to specify how the deviatoric tensor is related to the properties of the fluid. This called a constitutive equation and most non-Newtonian fluid models do this by relating the deviatoric stress to the rate of strain tensor $\dot{\gamma}_{ij}$, which we define as:

$$\dot{\gamma}_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right). \quad (1.5)$$

1.3 Non-Dimensional Groups in Non-Newtonian Fluid Mechanics

Dimensionless groups are useful for scaling arguments and for characterizing different flow regimes. For example, it is well known that a Taylor-Couette flow undergoes an ordered sequence of bifurcations during the transition to turbulence (Kerwell, 2005). For example, the different dynamical regimes (laminar flow, Taylor vortices, wavy Taylor vortices, and turbulence) can be characterized in terms of the Reynolds number based on the inner and outer diameter (Coles, 1965). The following discussion of important dimensionless quantities is strongly based on the article by Rob Poole in the British Society of Rheology Bulletin (Poole, 2012).

1.3.1 The Deborah Number

The Deborah number can be physically understood as the ratio of the relaxation time of the fluid to the time of observation. Thus, if λ denotes the relaxation time of the fluid and if T denotes a characteristic observation time, then we can define the Deborah number as:

$$De = \frac{\lambda}{T}. \quad (1.6)$$

Low Deborah numbers are indicative of more fluid-like behavior, whereas higher Deborah numbers mean that the material is closer to an elastic solid. We also note that for steady flows, the characteristic time of observation will be zero, so the Deborah number will be infinite, regardless of the relaxation time, so we need a another dimensionless group to characterize elasticity, which will be the Weissenberg number.

1.3.2 The Weissenberg Number

The Weissenberg Number can be interpreted as the ratio of elastic forces to viscous forces. If U represents a characteristic velocity scale and L a characteristic length scale, then we can define the Weissenberg Number as:

$$Wi = \frac{\lambda U}{L}. \quad (1.7)$$

In order to justify why the above ratio represents the ratio of elastic forces to viscous forces, we need to introduce the concept of a normal stress difference, which will be the dominant contribution to elastic forces. We will justify later why the first normal stress difference will be $2\lambda\mu\dot{\gamma}^2$. The viscous forces can be characterised by $\mu\dot{\gamma}$ and so the ratio of elastic to viscous ratio will be $2\lambda\dot{\gamma}$. Estimating $\dot{\gamma}$ as U/L recovers 1.7.

1.4 Normal Stress Differences

In a Newtonian fluid, the diagonal components of the stress tensor are all equal to minus the pressure. However, for a viscoelastic fluid, these components are not equal. We define the first normal stress difference N_1 , and the second normal stress difference N_2 as:

$$N_1 = \sigma_{xx} - \sigma_{yy} = \Psi_1 \dot{\gamma}^2 \quad (1.8)$$

$$N_2 = \sigma_{yy} - \sigma_{zz} = \Psi_2 \dot{\gamma}^2, \quad (1.9)$$

where Ψ_1 and Ψ_2 are the first and second normal stress difference coefficients, respectively. The first normal stress difference, when positive, can be thought of as a tension in the streamlines in the x-direction or as a pressure in the flow-gradient direction. Similarly, the second normal stress difference, when negative, can be thought of as tension in the vortex lines in the z direction or a pressure gradient against the flow gradient direction (Hinch, 2021).

The origin of these normal stresses can be explained using the microstructure of polymeric fluids. Polymers are stretched under the action of local shear and on average are aligned with the streamlines. The entropic forces in the polymer that try to return the molecule to its undisturbed conformation is what causes the tension in the streamlines. The normal stresses are the primary instigator of several well-known non-Newtonian phenomenology such as rod-climbing (Weissenberg effect) and the die swell effect. We will also mention that polymeric fluids have a positive N_1 and a N_2 so small that it is difficult to measure and thus is often neglected.

1.5 Standard Flows in Rheology

In order to determine a material's rheology, we need to experimentally probe the physical properties of the material. Experimental rheologists have recorded a significant amount of literature on non-Newtonian phenomenology organised around a small set of standard flows. This facilitates the comparison of experimental results. As pointed out by Morrison (Morrison, 2001), the motivation behind the choice of flow is twofold. Firstly, it must be relatively simple to describe mathematically, so that quantities such as the velocity field or the stress field can be readily calculated

from the experimental results. Secondly, the flow should be easily realisable experimentally.

1.5.1 Simple Shear Flow

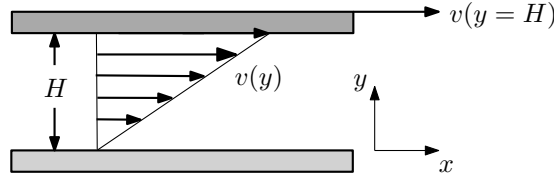


FIGURE 1.1: Flow field in simple shear

Shear flows are one of the most common flows in rheology, as they are easily experimentally realisable in the lab. As shown in figure 1.2, the flow consists of layers of fluids, that slide past each other and do not mix. The velocity field for simple shear is defined as:

$$\mathbf{u} = (\dot{\gamma}y, 0, 0).$$

The shear rate can be defined as $\dot{\gamma} = V/H$, where $V = v(y = H)$ is the speed of the top plate and H is the distance between the plates. The rate of strain tensor $\dot{\gamma}$ takes the following form:

$$\dot{\gamma} = \frac{1}{2} \begin{pmatrix} 0 & \dot{\gamma} & 0 \\ \dot{\gamma} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (1.10)$$

The deviatoric stress tensor of a Newtonian fluid in a shear flow $\tau_{ij} = 2\mu\gamma_{ij}$ reads:

$$\boldsymbol{\tau} = \mu \begin{pmatrix} 0 & \dot{\gamma} & 0 \\ \dot{\gamma} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

1.5.2 Extensional Flows

Extensional flow or elongational flows lead to diagonal components of the stress tensor. In this section, we will only describe uni-axial extensional flow, where the fluid gets stretched in one direction and compressed in the other two directions. The velocity field for simple shear is defined as:

$$\mathbf{u} = \dot{\epsilon}(x, -y/2, -z/2),$$

and the rate of strain tensor takes the following form:

$$\dot{\gamma} = \dot{\epsilon} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1/2 & 0 \\ 0 & 0 & -1/2 \end{pmatrix}, \quad (1.11)$$

where $\dot{\epsilon}$ is called the elongation rate and for uniaxial extensional flows, it is positive. This kind of flow arises in polymer processing operations such as fibre spinning.

For extensional flows, we can define a material function called an extensional viscosity, which characterises the resistance of the fluid to stretching motions. We will define the extensional viscosity μ_{ext} as:

$$\mu_{ext} = \frac{\sigma_{xx} - \frac{1}{2}\sigma_{yy} - \frac{1}{2}\sigma_{zz}}{3\dot{\epsilon}}. \quad (1.12)$$

1.6 Constitutive Relations

When modelling complex fluids, we account for how a particular material's microstructure influences its macroscopic properties by specifying a constitutive relation. For a Newtonian fluid, the constitutive law is particularly simple, as the deviatoric stress is linearly proportional to the rate of strain and the constant of proportionality is the shear viscosity μ . Mathematically, this is formulated as:

$$\sigma_{ij} = \mu \dot{\gamma}_{ij}. \quad (1.13)$$

However, for non-Newtonian fluids, the constitutive equations can get significantly more complicated, as the underlying constitutive model must reflect the intrinsic microstructure of the fluid or material. It is important to note that when modelling non-Newtonian fluid phenomena, there is often no perfect constitutive equation. Choosing a constitutive equation is both an art and a science, where one must isolate the key features of a material's response to deformation (such as the importance of shear-thinning or normal stresses) and then choose an appropriate constitutive equation that captures these characteristics. However, there are certain physical and mathematical constraints that constitutive relations must obey if they are to make sense physically. For example, the stress tensor must be symmetric and must be a tensor, so that the stress is independent of the coordinate system used to describe it. One of the most important mathematical constraints on a constitutive equation is that it must satisfy the principle of Frame Invariance, which is described in the next section.

1.6.1 The Principle of Frame Invariance

This is essentially the requirement that the physics, expressed by the constitutive equation, such as the response of a material to imposed stresses or to imposed deformation should be independent of the choice of reference frame. There are three parts to the local motion of a fluid element: a local velocity, a local rotation, and a local deformation. Thus, the principle of material frame invariance can be re-expressed as the constitutive equation should not depend on the translation, rotation or deformation of a frame of reference.

We can mathematically formulate the principle by using a Lagrangian description of the flow. Let X denote the Lagrangian coordinate and $x(X, t)$ the Eulerian coordinate. We define the deformation tensor as follows:

$$A_{ij} = \frac{\partial x_i}{\partial X_j}.$$

We assume that the system is local and causal. This means that the stress at a given material point depends only on the flow history of that material point and not its neighbours and that the stress cannot depend on future times. This means that $\sigma(t) = \sigma\{A(\tau)\}_{\tau \leq t}$.

Consider a change of frame of reference given by:

$$\mathbf{x}' = Q(t)\mathbf{x} + \mathbf{a}(t),$$

where $Q(t)$ is an orthogonal, rotational matrix and $\mathbf{a}(t)$ is a translation vector. We then require the stress in the new frame of reference σ' to satisfy:

$$\begin{aligned}\sigma' &= \sigma \{Q(\tau) A(\tau) Q(\tau)^T\}_{\tau \leq t} \\ &= Q(t) \sigma \{A(\tau)\}_{\tau \leq t} Q^T(t).\end{aligned}$$

This ensures that we would get the same result if we calculate our stress before or after rotating our frame of reference.

1.6.2 Time Derivative Problem

In the new reference frame, the stress σ' is given as:

$$\sigma' = Q \sigma Q^T. \quad (1.14)$$

Differentiating this we get:

$$\dot{\sigma}' = Q \dot{\sigma} Q^T + \dot{Q} \sigma Q^T + Q \sigma \dot{Q}^T. \quad (1.15)$$

Thus $\dot{\sigma}$ does not transform like a tensor, when we change a coordinate system. This is because when differentiating a tensor, we will have to differentiate both the components of the tensor and the changing basis vectors of the reference frame. In order to derive an objective time derivative, we will need to use a frame of reference moving with the fluid or material element that captures the local velocity, local rotation and local deformation. We introduce 3 time derivatives that are objective below. We do not prove that they satisfy the Principle of Frame Invariance and the reader is directed to Chapter 9 of the book by Morrison (Morrison, 2001) for a comprehensive proof.

The time derivative seen translating with local fluid velocity is the familiar material derivative:

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla. \quad (1.16)$$

The time derivative that is seen rotating with the material is the co-rotational time derivative, also called the Jaumann derivative. For a second-order tensor $\mathbf{A}$, it is defined as:

$$\overset{\circ}{\mathbf{A}} = \frac{D\mathbf{A}}{Dt} - \mathbf{A} \cdot \boldsymbol{\Omega} + \boldsymbol{\Omega} \cdot \mathbf{A}, \quad (1.17)$$

where $\boldsymbol{\Omega}$ is the vorticity tensor, defined as:

$$\Omega_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right). \quad (1.18)$$

If we require a time derivative that deforms with the fluid elements, then we have two possible choices, based on whether the components of the second-order tensor are covariant A_{ij} or contravariant A^{ij} . If the components are covariant, we get the lower-convective derivative defined as:

$$\overset{\Delta}{\mathbf{A}} = \frac{D\mathbf{A}}{Dt} + \nabla \mathbf{u} \cdot \mathbf{A} + \mathbf{A} \cdot (\nabla \mathbf{u})^T. \quad (1.19)$$

However, if the components of $\mathbf{A}$ are contravariant, A^{ij} , then we get the upper-convective derivative:

$$\overset{\nabla}{\mathbf{A}} = \frac{D\mathbf{A}}{Dt} - (\nabla \mathbf{u})^T \cdot \mathbf{A} - \mathbf{A} \cdot (\nabla \mathbf{u}). \quad (1.20)$$

We now explore some important and common constitutive relations.

1.7 Generalised Newtonian Fluid

This is essentially a generalisation of Newtonian fluids, where the viscosity depends on the shear rate $\dot{\gamma} = \sqrt{2\gamma_{ij}\dot{\gamma}_{ij}}$. The constitutive relation is:

$$\sigma_{ij} = -p\delta_{ij} + 2\mu(\dot{\gamma})\dot{\gamma}_{ij}. \quad (1.21)$$

These models were developed to fit experimental data and the form of $\mu(\dot{\gamma})$ is often derived empirically. We now explore some common forms of the functional relation of $\mu(\dot{\gamma})$.

1.7.1 Power-Law Fluids

The power-law model describes viscosity with a function that is proportional to some power of the shear rate:

$$\eta(\dot{\gamma}) = k\dot{\gamma}^{n-1}, \quad (1.22)$$

where k and n are fit parameters.

For $n < 1$, we get shear-thinning, where the shear stress ($\mu(\dot{\gamma})\dot{\gamma}$) decreases faster than linearly. This models the disintegration of microstructure under shear. Conversely, for $n > 1$, we get shear-thickening, where the shear stress increases faster than linearly, which implies the microstructure is built up by the fluid motion. Examples of shear-thickening liquids include corn flour and highly concentrated suspensions.

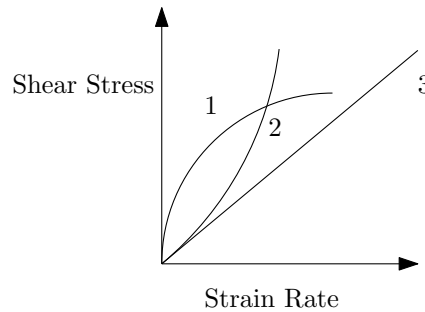


FIGURE 1.2: Power-Law fluid for 1) $n < 1$ shear-thinning fluid 2) $n > 1$ shear-thickening fluid and 3) $n = 1$ Newtonian Fluid

The Power-Law fluid has been extensively used for calculations in the polymer manufacturing process and the model has had considerable success in predicting flow-rate vs pressure drop data for processes such as extrusion, which involve high

shear rates (Morrison, 2001). However, the model has certain drawbacks. Firstly, it is strictly empirical and provides no molecular insight into any microstructure that could possibly influence the dynamics. Secondly, the Power-Law model for a shear-thinning fluid $n < 1$ predicts an infinite viscosity in the limit $\dot{\gamma} \rightarrow 0$.

1.7.2 Carreau-Yassuda Model

This is a generalisation of the shear-thinning power law fluid that regularises the infinite viscosity as $\dot{\gamma} \rightarrow 0$. It also captures more details of the shapes of experimentally measured curves of $\eta(\dot{\gamma})$ such as plateaus at low and high shear-rates. The model takes 5 parameters and takes the form:

$$\frac{\eta(\dot{\gamma}) - \eta_\infty}{\eta_0 - \eta_\infty} = (1 + (\dot{\gamma}\lambda))^{\frac{n-1}{a}}. \quad (1.23)$$

Notice that for $n < 1$, we have $\eta(\dot{\gamma}) \rightarrow \eta_\infty$ as $\dot{\gamma}\lambda \rightarrow +\infty$. Similarly, we have $\eta(\dot{\gamma}) \rightarrow \eta_0$ as $\dot{\gamma}\lambda \rightarrow 0$. Despite the success of the Carreau-Yasuda model in fitting most η vs $\dot{\gamma}$ curves, the model uses a relatively large number of parameters that have to be determined experimentally.

1.7.3 Yield Stress Fluids

Several non-Newtonian fluids exhibit a threshold stress, called a yield stress. This is a threshold stress (σ_y), such that for $\sigma < \sigma_y$, the material behaves like an elastic solid and for $\sigma > \sigma_y$, the material behaves like a fluid. Mud and lava are examples of yield-stress or viscoplastic fluids (Balmforth and Craster, 2001). If for $\sigma > \sigma_y$, the material behaves like a Newtonian fluid with constant viscosity, then we use the Bingham model:

$$\eta = \begin{cases} \infty & \sigma < \sigma_y \\ \mu_0 + \sigma_y/\dot{\gamma} & \sigma > \sigma_y \end{cases}$$

However, if for stresses above the yield stress, the material behaves like a power-law fluid, then we use the Hershel Bulkley model:

$$\eta = \begin{cases} \infty & \sigma < \sigma_y \\ \mu_0 \dot{\gamma}^{n-1} + \sigma_y/\dot{\gamma} & \sigma > \sigma_y \end{cases}$$

1.8 Linear Viscoelasticity

A linear viscoelastic fluid is a fluid that has a linear relationship between its strain history and its current stress value. This can be expressed in integral form as follows:

$$\tau(t) = \int_{-\infty}^t G(t-t') \dot{\gamma}(t') dt', \quad (1.24)$$

where τ is the shear stress, $\dot{\gamma}$ is the shear-rate and $G(t)$ is the fluid relaxation modulus. For a Newtonian fluid, $G(s) = 2\eta\delta(s)$.

1.9 Upper Convected Maxwell (UCM) Model

Viscoelastic liquids often exhibit characteristics of both viscous fluids and elastic solids. The Maxwell Model is a direct combination of the constitutive equations for

a Hookean elastic solid and a Newtonian Viscous Liquid. The constitutive equation for a Hookean elastic solid is:

$$\tau = G \gamma, \quad (1.25)$$

i.e., the shear stress τ is directly proportional to the shear strain, γ with the constant of proportionality called the elastic modulus, G .

The Hookean solid constitutive equation is a good model for rubber and metals at very small strains. The Maxwell model incorporates both elastic and viscous effects, and in its scalar form reads:

$$\tau + \frac{\eta}{G} \frac{\partial \tau}{\partial t} = 2\eta \dot{\gamma}. \quad (1.26)$$

As we approach a steady-state, the Maxwell model reduces to the constitutive equation for a Newtonian viscous fluid. For small times, the time derivative is dominant, so we retrieve the constitutive equation for a Hookean elastic solid. To convert the UCM model to a tensorial form, we replace the time-derivative with the upper-convective derivative to get:

$$\sigma + \frac{\eta}{G} \overset{\nabla}{\sigma} = 2\eta \dot{\gamma}. \quad (1.27)$$

1.10 Oldroyd-B Model

The Oldroyd-B model is a refinement of the UCM Model that has been extensively used to model polymer solutions. The stress tensor is divided into a polymeric and Newtonian solvent contribution: $\tau = \tau_p + \tau_s$ with a solvent contribution $\tau_s = 2\eta_s \dot{\gamma}$ and τ_p satisfying the UCM constitutive equation:

$$\tau_p + \lambda \overset{\nabla}{\tau}_p = 2\eta \dot{\gamma}, \quad (1.28)$$

where $\lambda = \eta/G$ is the relaxation time.

The Oldroyd-B constitutive equation can also be derived using ideas from Kinetic Theory, where a dilute solution of polymer molecules is modelled as a suspension of non-interacting Hookean dumbbell in a Newtonian solvent. Each polymer molecule is modeled as a dumbbell consisting of two beads connected by an infinitely extensible spring, which accounts for the drag experienced by the polymer molecule (Pearson, 1978).

For simple shear flow, the Oldroyd-B model predicts a shear-rate independent viscosity and a first normal stress coefficient of $\Psi = N_1/\dot{\gamma}^2$ (with N_1 denoting the first normal stress coefficient) and predicts a zero second normal stress difference ($N_2=0$). One of the shortcomings of the Oldroyd-B model is that in steady purely extensional (straining) flows, the stress and extensional viscosity diverge at a finite strain rate. This deficiency can be traced to the infinite extensibility of the Hookean dumbbells without bound. One way to rectify this error is to replace the Hookean spring by a non-linear spring, where the spring force diverges at maximum extension, which results in the 'FENE' (Finitely Extensible Nonlinear Elastic) model (Warner, 1972).

Chapter 2

Large Amplitude Oscillatory Shear (LAOS)

2.1 Small Amplitude Oscillatory Shear (SAOS)

The elastic and viscous characterisation of an unknown material can be examined simultaneously by imposing an oscillatory shear strain $\gamma(t) = \gamma_0 \sin(\omega t)$, which imposes an oscillatory shear strain rate $\dot{\gamma}(t) = \gamma_0 \omega \cos(\omega t)$ with γ_0 and ω being the amplitude and frequency of oscillation respectively.

The material sample is placed in a cone and plate or Taylor-Couette rheometer and a shear flow is set up. We assume that the amplitude is sufficiently small to ensure that the material is kept within its linear regime. Using the generalised linear viscoelastic fluid model we have:

$$\sigma(t) = \int_{-\infty}^t G(t-t') \dot{\gamma}(t') dt' \quad (2.1)$$

$$= \gamma_0 \omega \int_0^{\infty} G(s) \cos(\omega(t-s)) ds \quad (2.2)$$

$$= \gamma_0 \omega \Re \left[\exp(i\omega t) \int_0^{\infty} G(s) \exp(-i\omega s) ds \right], \quad (2.3)$$

where in the penultimate step we use the substitution $s = t - t'$ and in the final step we write $\cos(\omega(t-s)) = \Re[\exp(i\omega(t-s))]$.

We now define the complex shear modulus G^* as:

$$G^* = i\omega \int_0^{\infty} G(s) \exp(-i\omega s) ds \quad (2.4)$$

with real and imaginary parts $G^* = G' + iG''$, with G' called the storage modulus and G'' called the loss modulus. Then 2.3 can be re-expressed as:

$$\sigma = -i\gamma_0 \Re[\exp(i\omega t) G^*]. \quad (2.5)$$

By writing $\exp(i\omega t) = \cos(\omega t) + i \sin(\omega t)$ in 2.5 and using the expressions $\gamma(t) = \gamma_0 \sin(\omega t)$ and $\dot{\gamma}(t) = \gamma_0 \omega \cos(\omega t)$ we get:

$$\sigma(t) = G' \gamma(t) + \frac{G''}{\omega} \dot{\gamma}(t). \quad (2.6)$$

By splitting the shear stress in this way, we see that there is a component of the stress that is in phase with the imposed strain and a component that is in phase with the imposed strain rate. For a Newtonian fluid in SAOS, the shear stress response is in phase with the strain rate $G' = 0$ and $\eta = G''/2\omega$. Similarly for a Hookean elastic

solid, $G'' = 0$ and $G' = G_0$, which is the elastic shear modulus. Thus G' is called the storage modulus, because it measures the material's ability to store elastic energy. G'' is called the loss modulus, as it is related to viscous part of the stress response and is related to the dissipation of energy.

2.1.1 Lissajous-Bowditch Curves

For a linear viscoelastic fluid, subject to an imposed oscillatory strain, $\gamma = \gamma_0 \sin(\omega t)$, will also produce a sinusoidal stress response delayed by a phase angle δ :

$$\sigma(t) = \sigma_0 \sin(\omega t + \delta). \quad (2.7)$$

Using the trig-addition formula, we have:

$$\sigma = \sigma_0 \cos(\delta) \sin(\omega t) + \sigma_0 \sin(\delta) \cos(\omega t), \quad (2.8)$$

$$= \frac{\sigma_0}{\gamma_0} \cos(\delta) \gamma(t) + \frac{\sigma_0}{\dot{\gamma}_0} \sin(\delta) \dot{\gamma}(t), \quad (2.9)$$

where $\dot{\gamma}_0 = \gamma_0 \omega$.

Comparing the above expression with equation 2.6, we get:

$$G' = \frac{\sigma_0}{\gamma_0} \cos \delta \quad (2.10)$$

$$G'' = \frac{\sigma_0}{\dot{\gamma}_0} \sin \delta. \quad (2.11)$$

The phase angle, δ and the amplitude of the stress response, σ_0 , can be easily obtained by plotting the stress response against the imposed strain. These plots are called Lissajous-Bowditch plots and provide a useful graphical approach to describing the viscoelasticity of a fluid. For a linearly elastic (Hookean) solid, $\sigma = G_0 \gamma$, so the Lissajous curve will be a straight line through the origin. For a purely viscous liquid, $\sigma = 2\eta \dot{\gamma} = 2\eta \dot{\gamma}_0 \sin(\omega t + \pi/2)$. This corresponds to an upright (not slanted) ellipse or a circle centered at the origin. For a viscoelastic material subjected to sinusoidal strain, the Lissajous curves are slanted ellipses centered at the origin.

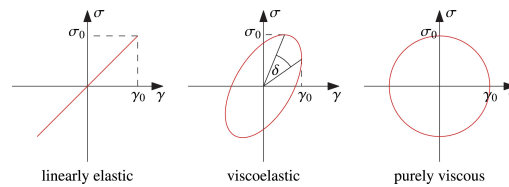


FIGURE 2.1: Lissajous curves for a linearly elastic solid, viscoelastic fluid, and purely viscous fluid. Source: Townsend and Wilson, 2018

2.2 Large Amplitude Oscillatory Shear

When the strain amplitude is large, we enter a non-linear regime, called Large Amplitude Oscillatory Shear (LAOS), where the stress curve as a function of time cannot be described by a single trigonometric function. An example of a non-linear stress response is shown in Figure 2.3.

There are myriad applications of LAOS experiments. There is evidence (Yosick and Giacomini, 1996) that these experiments can detect subtle differences in the molecular structure of plastics, which can in turn have a profound effect on their processability. In addition, LAOS experiments can also be used to characterise polymer long chain branching (Dutta et al., 2024). A parameter known as the polymer long chain branching index, can be quantitatively obtained through an empirical equation that uses LAOS higher harmonic signals from the third and fifth harmonic. This is particularly useful for polymer processing operations in industry, as branching can affect the rheological properties of polymer melts, such as relaxation times and extensional viscosities.

Furthermore, in this non-linear regime, the storage and loss moduli are no longer valid as measures of the elastic and viscous characterisations of the stress response, as they are only defined within the linear viscoelastic limit and thus lose their physical meaning in the LAOS regime. One common method for interpreting LAOS data is the Fourier Transform Method.

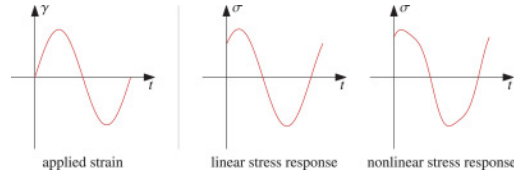


FIGURE 2.2: A linear and non-linear stress response

2.2.1 Fourier Transform Rheology

The Fourier Transform approach is based on representing the non-linear stress response in LAOS as a Fourier Series, which allows us to include higher harmonics, which contribute to the non-linearity of the stress response. Only odd harmonics are required due to the odd symmetry of the stress response ($\sigma(-\gamma) = -\sigma(\gamma)$):

$$\sigma(t; \omega, \gamma_0) = \gamma_0 \sum_{n: \text{odd}} \{G'_n(\omega, \gamma_0) \sin n\omega t + G''_n(\omega, \gamma_0) \cos n\omega t\}. \quad (2.12)$$

Despite the ease at which the non-linear stress signals of two materials can be compared using the Fourier Transform, the method does possess its own set of drawbacks. Firstly, the Fourier Transform does not provide a clear physical interpretation of higher order coefficients. Secondly, the use of the storage and loss moduli in the non-linear regime often fails to identify or distinguish important microstructural features. As pointed out by Ewoldt (Ewoldt, Hosoi, and McKinley, 2008), there have been several paradoxical observations of soft biological materials, where strain-stiffening is suggested by the elastic Lissajous-Bowditch curves, even though the storage modulus does not appear to increase as a function of strain amplitude.

2.3 Non-Linear Viscoelastic Moduli

We now describe the framework introduced by Ewoldt (Ewoldt, Hosoi, and McKinley, 2008) to interpret the stress response from LAOS experiments. The framework is based on an earlier idea by Cho (Cho et al., 2005) to express the stress response using an orthogonal Chebyshev basis in the deformation domain. We first express the

stress response as a function of the strain and strain rate by introducing the variables x and y defined below:

$$x = \gamma/\gamma_0 = \sin(\omega t), \quad (2.13)$$

$$y = \dot{\gamma}/\dot{\gamma}_0 = \cos(\omega t), \quad (2.14)$$

$$\sigma(\omega, \gamma_0, t) = \sigma(x, y). \quad (2.15)$$

The total oscillatory stress is decomposed into the sum of two components: an elastic stress σ' and a viscous stress σ'' as follows:

$$\sigma(t) = \sigma'(t) + \sigma''(t).$$

This follows from the fact that the stress is odd in both x and y , so $\sigma(-x, -y) = -\sigma(x, y)$. This leads to the following decomposition for stress:

$$\sigma(x, y) = \underbrace{\frac{\sigma(x, y) - \sigma(-x, y)}{2}}_{\text{odd for } x \text{ and even for } y} + \underbrace{\frac{\sigma(x, y) - \sigma(x, -y)}{2}}_{\text{even for } x \text{ and odd for } y}. \quad (2.16)$$

The elastic (σ') and viscous (σ'') components of stress are:

$$\sigma' = \frac{\sigma(x, y) - \sigma(-x, y)}{2} = \gamma_0 \sum_{n \text{ odd}} G'_n \sin(n\omega t) \quad (2.17)$$

$$\sigma'' = \frac{\sigma(x, y) - \sigma(x, -y)}{2} = \gamma_0 \sum_{n \text{ odd}} G''_n \cos(n\omega t) \quad (2.18)$$

We can decompose the elastic and viscous stress in terms of the orthogonal Chebyshev polynomials:

$$\sigma^e(x; \omega, \gamma) = \gamma_0 \sum_{n: \text{ odd}} e_n(\omega, \gamma_0) T_n(x), \quad (2.19)$$

$$\sigma^v(y; \omega, \dot{\gamma}) = \dot{\gamma}_0 \sum_{n: \text{ odd}} v_n(\omega, \gamma_0) T_n(y), \quad (2.20)$$

where $T_n(x)$ is the n th order Chebyshev polynomial of the first kind. As pointed out by Ewoldt (Ewoldt, Hosoi, and McKinley, 2008), the orthogonality of the Chebyshev polynomials over the finite domain $[-1, 1]$ ensures that the Chebyshev coefficients are unique, so the higher order terms do not affect the lower order terms. Secondly, the choice of the Chebyshev polynomials over other set of orthogonal polynomials ensures that the Chebyshev and Fourier coefficients are simply related:

$$e_n = G'_n(-1)^{(n-1)/2}, \quad (2.21)$$

$$v_n = \frac{G''_n}{\omega}. \quad (2.22)$$

2.3.1 Elastic Modulus

We first note that the storage modulus can be a misleading measure of the elastic characterisation of a material's stress response, as other harmonics can also store

energy. The storage modulus can be more precisely defined as a measure of the average elasticity in the material response. To capture the local behaviour of the elastic response, we define a set of material functions.

Minimum Strain Modulus

G'_M is the minimum strain modulus or tangent modulus at $\gamma = 0$.

$$G'_M = \left. \frac{d\sigma}{d\gamma} \right|_{\gamma=0} = \sum_{n \text{ odd}} n G'_n = e_1 - 3e_3 + 5e_5 + \dots$$

At zero strain, the strain rate is a maximum and thus is locally constant. Hence the viscous contribution to the stress is locally constant at this point.

Large Strain Modulus

G'_L is the large strain modulus or secant modulus evaluated at maximum imposed strain.

$$G'_L = \left. \frac{\sigma}{\gamma} \right|_{\gamma=\pm\gamma_0} = \sum_{n \text{ odd}} (-1)^{(n-1)/2} G'_n = e_1 + e_3 + e_5 + \dots$$

Notice that at zero strain rate, the viscous contribution to the stress at this point locally vanishes.

2.3.2 Dynamic Viscosity

As with the elastic modulus, we can introduce η'_M and η'_L , which represent the instantaneous viscosity at the smallest and largest shear-rates, respectively.

Minimum Rate Dynamic Viscosity

$$\eta'_M = \left. \frac{d\sigma}{d\dot{\gamma}} \right|_{\dot{\gamma}=0} = \frac{1}{\omega} \sum_{n \text{ odd}} n G''_n = v_1 - 3v_3 + 5v_5 + \dots$$

Large Rate Dynamic Viscosity

$$\eta'_L = \left. \frac{\sigma}{\dot{\gamma}} \right|_{\dot{\gamma}=\pm\dot{\gamma}_0} = \frac{1}{\omega} \sum_{n \text{ odd}} (-1)^{(n-1)/2} G''_n = v_1 + v_3 + v_5 + \dots$$

2.3.3 Dimensionless Indices of Non-Linearity

Strain-Stiffening Index

If the large-strain modulus is greater than the minimum strain modulus ($G'_L > G'_M$) within a particular cycle, then the response is strain-stiffening. This motivates the following definition for the Strain-Stiffening ratio:

$$S = \frac{G'_L - G'_M}{G'_L} = \frac{4(e_3 - e_5 + \dots)}{e_1 + e_3 + e_5 + \dots}. \quad (2.23)$$

$$S > 0 \implies \text{Intracycle Strain Stiffening}$$

$$S < 0 \implies \text{Intracycle Strain Softening}$$

Tension-Thickening Index

We can analogously define a tension-thickening ratio as:

$$T = \frac{\eta'_L - \eta'_M}{\eta'_L} = \frac{4(v_3 - v_5 + \dots)}{v_1 + v_3 + v_5 + \dots}.$$

$T > 0 \implies$ Intracycle Shear-Thickening

$T < 0 \implies$ Intracycle Shear-Thinning

The geometric interpretation of these non-linear material functions is illustrated in figure, which has been taken from the original paper by Ewoldt (Ewoldt, Hosoi, and McKinley, 2008).

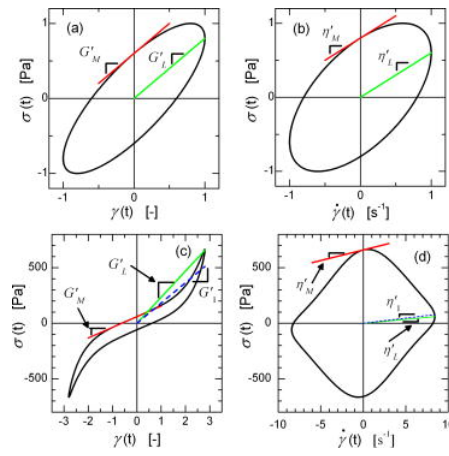


FIGURE 2.3: Geometric interpretation of the non-linear material function

Chapter 3

A New Model for LAOS Data

3.1 A Conjecture by Dr. Peter Laity

A new empirical model by Dr. Peter Laity (unpublished at the time of writing) suggests modelling the stress response of a viscoelastic polymer solution and elastic gel from a LAOS experiment in the form:

$$\sigma(t) = A_1 \sin(\omega t + \delta_1) + A_2 \sin^5(\omega t + \delta_2), \quad (3.1)$$

where A_1 , A_2 , δ_1 and δ_2 are the 4 adjustable parameters of the model.

Despite the lack of a theoretical basis for the model, they claim it provides a reasonable fit to the stress response of native silk feedstock (NSF) under LAOS experiments. Plots of various elastic Lissajous curves in Figure 3.1 illustrate how we approach the linear viscoelastic regime when the linear term is significantly larger than the non-linear term.

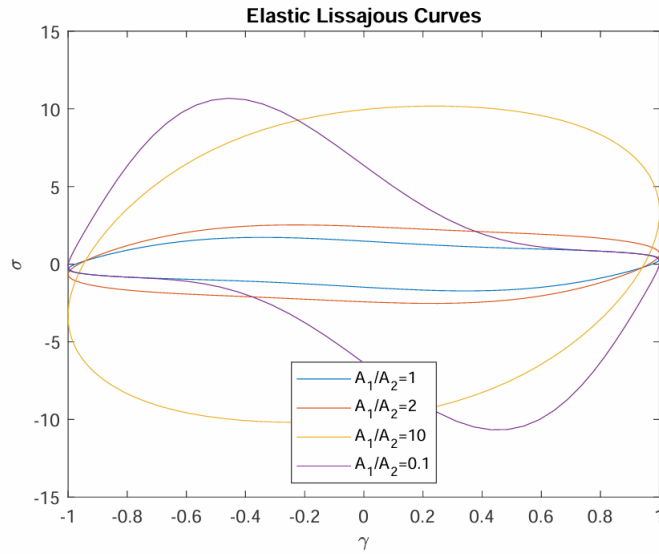


FIGURE 3.1: Elastic Lissajous Curves at different amplitudes

3.2 Investigating the Stress Response

We now investigate the rheological properties of the viscoelastic materials that would give rise to the kind of stress response that has been proposed by Dr. Peter Laity in equation 3.1. We now analyse the non-linear stress response under the lens of the

framework introduced in Chapter 2. We start by calculating all of the non-linear material functions introduced in the previous section, by expressing the stress response in a Chebyshev basis instead of the conventional Fourier or sinusoidal basis.

3.2.1 Chebyshev Coefficients

$$e_1 = \frac{1}{\gamma_0} (A_1 \cos(\delta_1) + \frac{5}{8} A_2 \cos \delta_2) \quad (3.2)$$

$$e_3 = \frac{5A_2}{16\gamma_0} \cos(3\delta_2) \quad (3.3)$$

$$e_5 = \frac{A_2}{16\gamma_0} \cos(5\delta_2) \quad (3.4)$$

$$v_1 = \frac{1}{\omega\gamma_0} (A_1 \sin(\delta_1) + \frac{5}{8} A_2 \sin(\delta_2)) \quad (3.5)$$

$$v_3 = -\frac{5}{16\omega\gamma_0} A_2 \sin(3\delta_2) \quad (3.6)$$

$$v_5 = \frac{1}{16\omega\gamma_0} A_2 \sin(5\delta_2) \quad (3.7)$$

3.2.2 Elastic Modulus

$$G'_M = \frac{A_1}{\gamma_0} \cos \delta_1 + \frac{5A_2}{8\gamma_0} \cos \delta_2 - \frac{15A_2}{16\gamma_0} \cos(3\delta_2) + \frac{5A_2}{16\gamma_0} \cos(5\delta_2) \quad (3.8)$$

$$G'_L = \frac{A_1}{\gamma_0} \cos \delta_1 + \frac{A_2}{\gamma_0} \cos^5 \delta_2 \quad (3.9)$$

3.2.3 Strain-Stiffening Ratio

$$S = \frac{A_2(5 \cos 3\delta_2 - \cos 5\delta_2)}{4(A_1 \cos \delta_1 + A_2 \cos^5 \delta_2)} \quad (3.10)$$

3.2.4 Dynamic Viscosity

$$\eta_M = \frac{1}{\omega\gamma_0} (A_1 \sin \delta_1 + \frac{5}{8} A_2 \sin \delta_2 + \frac{15}{16} A_2 \sin 3\delta_2 + \frac{5A_2}{16} \sin 5\delta_2) \quad (3.11)$$

$$\eta_L = \frac{A_1}{\omega\gamma_0} \sin \delta_1 + \frac{A_2}{\omega\gamma_0} \sin^5 \delta_2 \quad (3.12)$$

3.2.5 Tension-Thickening Index

$$T = -\frac{A_2(5 \sin 3\delta_2 + \sin 5\delta_2)}{4(A_1 \sin \delta_1 + A_2 \sin^5 \delta_2)} \quad (3.13)$$

$$(3.14)$$

$$= -\frac{(5 \sin 3\delta_2 + \sin 5\delta_2)}{4((B + \sin^5 \delta_2))} \quad (3.15)$$

$$B = \frac{A_1}{A_2} \sin \delta_1 \quad (3.16)$$

3.2.6 A Further look at the Tension-Thickening Index

To investigate how the phase angles δ_1 and δ_2 affect the physical properties of the material, we first fix $\delta_2 = 117.8^\circ$ and then plot the tension thickening index T from 3.15 against δ_1 for different values of the amplitude ratio B . The results are shown in figure 3.2. Next, we fix $\delta_1 = 70.3^\circ$ and then plot the graph of T against δ_2 for different values of B as before, as shown in figure 3.3.

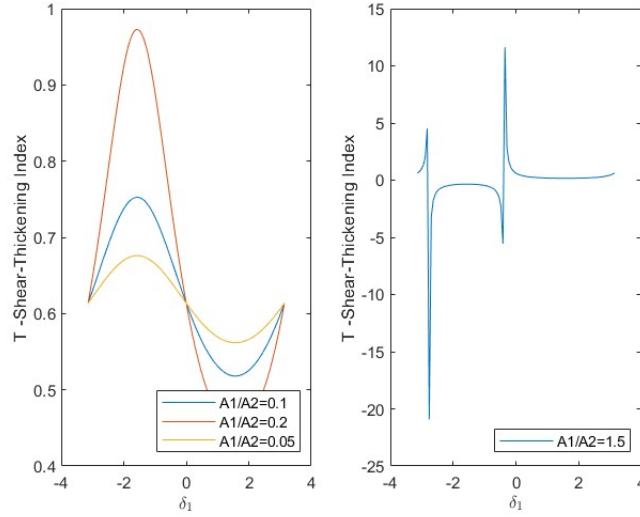


FIGURE 3.2: The Tension Thickening Index against δ_1

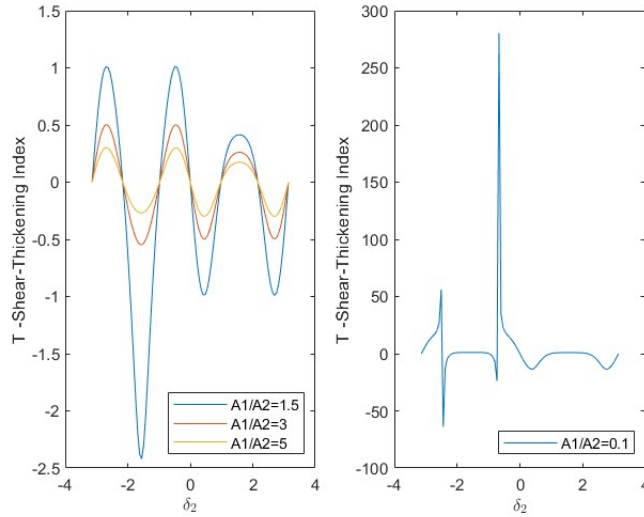


FIGURE 3.3: The Tension Thickening Index against δ_2

As seen in both figures 3.2 and 3.3, the Tension-Thickening index is unbounded at a finite value of the phase angles for either sufficiently large (if δ_2 is fixed) or small values (if δ_1 is fixed) of the ratio A_1/A_2 . This can be explained in terms of equation 3.15. In the case where δ_1 is fixed, the denominator of equation 3.15 has a root if $\left| \frac{A_1}{A_2} \sin(\delta_1) \right| \leq 1$, which occurs for sufficiently small values of the ratio A_1/A_2 .

Analogously when δ_2 is fixed, the same denominator has a root, when $\left| \frac{A_2}{A_1} \sin^5(\delta_2) \right| \leq 1$, which occurs for sufficiently large values of A_1/A_2 .

Furthermore, it is well-known that the native silk feedstock, that was used in the original, unpublished study exhibits characteristics of polymeric liquids (Laity and Holland, 2017). However, it can be clearly seen in figures 3.2 and 3.3 that there are values of δ_1 and δ_2 , where the Tension-Thickening index is positive, which suggests shear-thickening. This contradicts the well-known shear-thinning nature of polymeric fluids (Pearson, 1978). Overall, this suggests that there are certain limitations of the Tension-Thickening index.

In order to further investigate this discrepancy, we will calculate a set of values for the Tension-Thickening index, based on the set of values for the model parameters $\delta_1, \delta_2, A_1, A_2$ that were used to fit the experimental data from Dr. Peter Laity's study. The data was extracted from the unpublished study by Dr. Peter Laity. Three different samples with different shear viscosities, η , were used and for each sample, 6 different measurements were taken. The results are shown in table 3.1. As before, there seem to be a considerable number of readings that have a positive value of T , which would suggest shear-thickening.

η (Pa s)	Shear Thickening Index (T)					
780	-0.0766	-0.0094	-0.0739	-0.2610	-0.3071	-0.3416
3429	-0.0817	0.0372	0.2448	0.4399	-0.5950	-0.6280
6571	-0.0205	0.0845	-0.0401	-0.2231	-0.1301	-1.0214

TABLE 3.1: Shear-Thickening Index at different strain amplitudes

Such discrepancies in these non-linear material functions have been previously documented in the literature. For example, a study by Randy Ewoldt (Ewoldt and Bharadwaj, 2013) carried out experimental measurements of the non-linear material functions for polyvinyl alcohol (PVA) and sodium tetraborate (Borax). These non-linear material functions were identified for a range of frequencies. When comparing their experimental values with those predicted by the co-rotational maxwell model, they obtained a 'remarkable similarity' in the shapes and the normalised magnitudes. However all the signs of the non-linear material functions were opposite to those predicted by the co-rotational maxwell model. They also note that there is no adjustable parameter in the co-rotational maxwell model to change the sign of the response to match the experimental observations.

Chapter 4

Shear Banding

4.1 An Introduction to Shear-Banding

Shear banding is a transition between a homogeneous to a non-homogeneous state of flow, where the fluid separates into different regions (bands), which have different shear rates. For example, shear banding has been identified in worm-like micelles using flow birefringence (Makhloufi et al., 1995) and NMR (Nuclear Magnetic Resonance) velocimetry (P.T. Callaghan et al., 1996). However, it has been observed in many other complex fluids (Berret, n.d.), such as cubic phases of soft colloids, electro- and magneto-rheological fluids, transient networks of associative polymers and soft glassy materials.

Theoretically, shear-banding arises from constitutive equations that have a non-monotone relationship between shear stress and shear-rate (Renardy, 1995). One example of such a constitutive equation is the Johnson-Segalman equation (Johnson and Segalman, 1977), which includes a polymeric contribution to the stress: Σ :

$$\sigma = -p\mathbf{I} + 2\epsilon\dot{\gamma} + \Sigma \quad (4.1)$$

$$\overset{\Delta}{\Sigma} = 2\dot{\gamma} - \Sigma, \quad (4.2)$$

where ϵ is the retardation parameter and the Johnson-Segalman time derivative is defined as:

$$\overset{\Delta}{\Sigma} = \left(\frac{\partial}{\partial t} + U_k \partial_k \right) \Sigma_{ij} - [\Sigma_{ik} \Omega_{kj} - \Omega_{ik} \Sigma_{kj} + a(E_{ik} \Sigma_{kj} + \Sigma_{ik} E_{kj})], \quad (4.3)$$

with a being the slip parameter and E and Ω are the rate of strain tensors and vorticity tensors.

If we assume steady flow, we get the following set of equations from the constitutive equation:

$$\Sigma_{xy} - \frac{1}{2}(1-a)\dot{\gamma}\Sigma_{xx} + \frac{1}{2}(1+a)\dot{\gamma}\Sigma_{yy} = \dot{\gamma} \quad (4.4)$$

$$\Sigma_{xx} = -\dot{\gamma}(1+a)\Sigma_{xy} \quad (4.5)$$

$$\Sigma_{yy} = \dot{\gamma}(1-a)\Sigma_{xy}, \quad (4.6)$$

we can then solve for Σ_{xy} as function of $\dot{\gamma}$ as:

$$\Sigma_{xy} = \frac{\dot{\gamma}}{1 + \dot{\gamma}^2(1-a^2)}. \quad (4.7)$$

From the constitutive equation we have:

$$S_{xy} = \epsilon \dot{\gamma} + \frac{\dot{\gamma}}{1 + \dot{\gamma}^2(1 - a^2)} \quad (4.8)$$

We can plot the shear stress against the shear rate for $\epsilon = 0.05$ and $a = 0.8$ to illustrate the non-monotonic nature of the constitutive relation, as shown in figure 4.1.

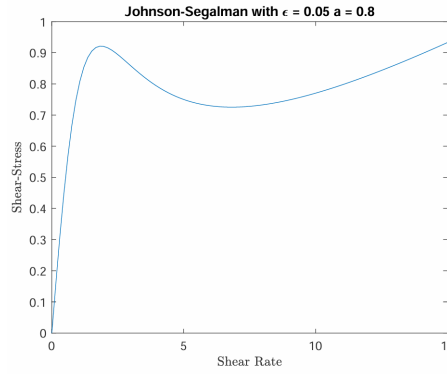


FIGURE 4.1: Shear Stress against Strain Rate Curve for $\epsilon = 0.05$ and $a = 0.8$

4.2 Evidence of Shear Banding

One of the main signatures of shear banding in complex fluids is a region in the steady state flow curve, where the stress remains nearly constant despite a change in shear rate. In Dr. Laity's unpublished paper, they graphically present some experimental data from Native Silk Feedstock (NSF) subject to LAOS measurements as an elastic Lissajous curve (4.2). We notice that for large strain amplitudes, the instantaneous stress at zero strain is unchanged despite changes in the strain amplitude and thus shear-strain amplitude.

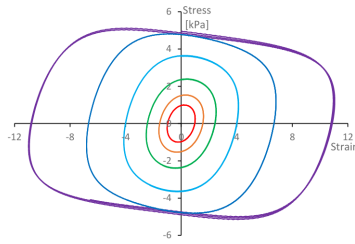


FIGURE 4.2: Stress response for a NSF specimen during LAOS measurements from Dr. Laity's unpublished work

4.3 Extensions

In order to probe this anomaly further, the natural subsequent step would be to perform a numerical investigation of shear-banding by numerically solving the momentum and continuity equation in conjunction with the Johnson-Segalman constitutive equation using a finite-difference scheme with appropriate boundary conditions. The analysis would be in a similar spirit to that of Wilson and Fielding, 2006, except that we are dealing with an unsteady problem. The aim would be to compare the Lissajous curve, generated from the numerical simulations to figure 3.1, which was based on experimental data. However, due to time constraints, we were unable to explore this avenue further.

Chapter 5

Conclusions

Viscoelastic liquids exhibit both viscous and elastic characteristics. Polymeric fluids are a ubiquitous example of viscoelastic fluids in industry, as they can store elastic energy when molecules stretch and rotate as a result of the flow and entropic forces act to bring the polymer chains to their equilibrium configuration. When modelling viscoelastic fluids, we need specify a constitutive equation, which relates the stress tensor to the rate of strain tensor.

Both the viscous and elastic characteristics of an unknown material can be examined simultaneously using a small amplitude oscillatory shear (SAOS). At small strain amplitudes, so the material is within its linear regime, the material can be characterised by the storage and loss moduli, which are the components of the stress response, that in phase with the shear and the shear rate, respectively. Large amplitude oscillatory shear (LAOS) experiments are an extension of SAOS, where large strain amplitude, where larger strain amplitudes are imposed resulting in the appearance of higher harmonics and non-linear stress response.

The framework pioneered by Ewoldt, enables us to exploit the rich non-linearities of the stress response to infer the physical properties of the sample through the use of the non-linear material functions that we have discussed here. Subsequently, we analyse the model for the stress response, introduced by Dr. Peter Laity using the lens of the Ewoldt framework. Interestingly, we encounter discrepancies in the predictions made by the non-linear material functions. Namely, we discover that the shear-thickening index surprisingly suggests that the polymeric fluids used by Dr. Peter Laity are shear-thickening. As discussed in Chapter 3, these issues have already been documented by previous studies and our study reinforces the shortcomings of the Ewoldt framework, when analysing empirical data, that must be further investigated and addressed.

Although we have not presented any tangible conclusions from our study, we do hope that our investigation can serve as the starting point for a more detailed analysis of the physical properties that would drive the stress response that was devised by Dr. Laity.

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